

Lecture 11 Fractional Helly &

Kruskal - Katona

Thm 1.1 (Frac. Helly) Let $\alpha \in (0, 1)$, $\mathcal{F}$ col^o of conv. sets in $\mathbb{R}^d$

If $|\mathcal{F}| \geq \alpha \binom{|\mathcal{F}|}{d+1}$ $(d+1)$ -tuples $A \in \binom{\mathcal{F}}{d+1}$ satisfy $\cap A \neq \emptyset$, then

$$\exists \mathcal{F}' \subseteq \mathcal{F} \quad \text{s.t.} \quad \begin{cases} \cap \mathcal{F}' \neq \emptyset \\ |\mathcal{F}'| \geq \beta |\mathcal{F}| \end{cases} \quad \text{where } \beta \geq 1 - (1-\alpha)^{1/(d+1)}$$

$|\mathcal{F}| = n$

Upper bound Thm by Alon - Kalai

Recall A **simplicial complex** K is a set of simplices

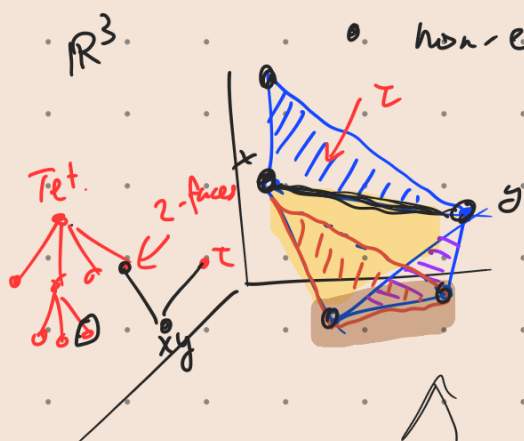
s.t.
 • Every face of a simplex in K also in K

• non-empty intersection of $\sigma_1, \sigma_2 \in K$ is a face of both σ_1, σ_2 .

(xy is not free)

Notation $f_j(K) = \# j\text{-dim. faces of } K$

$$f_0(K) = 5$$



Def A face $\sigma \in K$ is a **free face** if it is

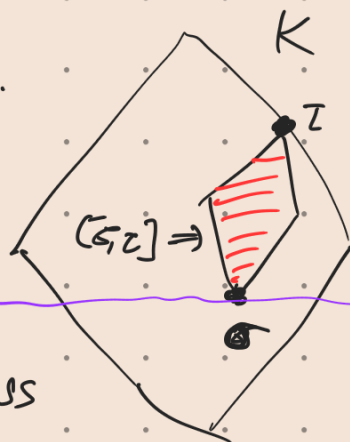
contained in a unique maximal face τ .

Given σ free, $\tau \supseteq \sigma$ is the unique max

write $[\sigma, \tau] = \{\text{all faces } \supseteq \sigma\}$

An **elementary (a,b)-collapse** is the process

of removing $[\sigma, \tau]$ for some free $|\sigma| = a, |\tau| = b$



Def A Simplicial complex is d -collapsible if all faces of size $\geq d$ can be removed after some seq of elementary collapses, in which, each collapse is of type (d, d') for some $d' \geq d$.

Ex: a tree is 1-collapsible. (a free face of size = leaf)

i.e. $K \rightarrow K_1 = K \setminus [\sigma_1, \tau_1] \rightarrow K_2 = K_1 \setminus [\sigma_2, \tau_2] \rightarrow \dots \rightarrow K_t$, where $f_{d-1}(K_t) = 0$

Thm (AK 85) Let K be a d -collapsible sim. complex on n vxs. If $\dim K < d+r$, i.e. $f_{d+r}(K) = 0$,

$$\Rightarrow \forall d \leq j \leq d+r-1, \quad f_j(K) \leq \sum_{i=j+1-d}^r \binom{n-i}{d} \binom{i-1}{j-d}$$



In particular,

$$f_d(K) \leq \binom{n}{d+1} - \binom{n-r}{d+1} \dots (*)$$

Ex Derive frac. Helly from (*). Take $r = \beta n - d$

Apply the above to nerve complex

Def: The nerve complex of $\mathcal{F}$ (colⁿ conv. sets in $\mathbb{R}^d$)

vx = elem^t $F \in \mathcal{F}$

face = tuple of sets $\{F_1, \dots, F_i\} \in \mathcal{F}$ that has non-empty intersection.

Pf: Fix a seq

$$K \rightarrow K_1 = K \setminus [\sigma_1, \tau_1] \rightarrow K_2 = K_1 \setminus [\sigma_2, \tau_2] \rightarrow \dots \rightarrow K_t, \text{ where } f_{d-r}(K_t) = 0$$

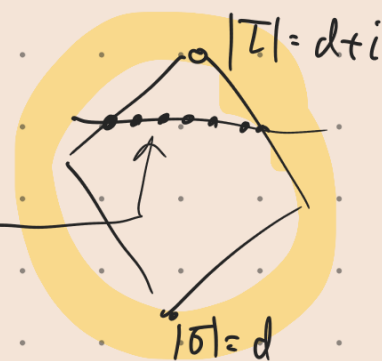
where $|\sigma_i| = d$, $|\tau_i| \geq d$

• For each $i \geq 0$, write $h_i = \# \text{ ele}^g(d, d+i) \text{ - collapses}$.

• Note in each such $(d, d+i)$ collapse, $\#$ j -dim faces deleted

$$d+1 \leq j \leq d+i-1, \quad = \binom{i}{j+1-d}$$

Out of $|\tau \setminus \sigma| = i$ choose $j+1-d$ elements to add to σ to get a j -dim face.



• Recall that $f_{d+r}(K) = 0$ for $0 \leq i \leq r$,

write $H_i = \sum_{i'=i}^r h_{i'} \quad (\# \text{ of } (d, \geq d+i) \text{ - collapse})$

Then for each $d \leq j \leq d+r-1$,

$$\begin{aligned} f_j(K) &= \sum_{i=j+1-d}^r h_i \binom{i}{j+1-d} = \sum (H_i - H_{i+1}) \binom{i}{j+1-d} \\ &= \sum_{i=j+1-d}^r H_i \cdot \binom{i-1}{j-d} \end{aligned}$$

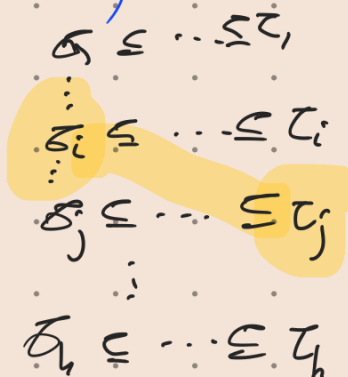
• Left to show $\forall i \geq 0$,

$$H_i \leq \binom{n-i}{d}$$

$\hookrightarrow \# \text{ of } (d, d') \text{ - collapses, } d' \geq d+i$.

Let $[\sigma_k, \tau_k]$, $k \in [h]$, be the collapses corresponding to H_i . ($|\tau_k| \geq d+i$)

- $\forall i \in [h], \sigma_i \in \tau_i \Rightarrow A_i \cap B_i = \emptyset$
- $\forall i < j, \sigma_i \notin \tau_j \Rightarrow A_i \cap B_j \neq \emptyset$



Let $A_i = \sigma_i$, $B_i = [n] \setminus \tau_i$

$$|A_i| = d, \quad |B_i| \leq n - d - i$$

Skewed set-pair inequality $\Rightarrow h = H_i \leq \binom{d+n-d-i}{d} \leq \binom{n-i}{d}$

Kruskal - Katona

- Erdős - Rademacher Δ minimisation (40.5) ← notoriously difficult

Given n -vc & m edges $\Rightarrow$ min. $\neq \Delta$?

↑
max ?

much easier

consequence of Kruskal - Katona thm.

Ex Solve Δ -max. problem using K.K.

Def $\mathcal{F} \subseteq \binom{[n]}{k}$ its shadow $\partial \mathcal{F} \subseteq \binom{[n]}{k-1}$

consisting of all $(k-1)$ -tuples contained in some edge of $\mathcal{F}$.

$$\partial \mathcal{F} = \left\{ F \in \binom{[n]}{k-1} : F \subseteq G, G \in \mathcal{F} \right\}$$

- K.K. studies size of shadow.

Q: $n \geq k$, $0 \leq m \leq \binom{n}{k}$ What is the min. size shadow of an n -x k -unif $\mathcal{F}$ with m edges?

Colex

Try $k=3$, 3-unif hypergraph $\mathcal{F}$, ($\partial\mathcal{F}$ a graph)

- Fix $n_3 \in \mathbb{N}$ s.t. $\binom{n_3}{3} \leq m < \binom{n_3+1}{3}$

Fix n_2 s.t. Take $|V_3| = n_3$

- $\binom{n_2}{2} \leq m - \binom{n_3}{3}$ edges left $< \binom{n_2+1}{2}$

Take all triples $\{v, e\}$, $e \in \binom{V_2}{2}$
 $|V_2| \leq V_3$

$\parallel$
 n_2

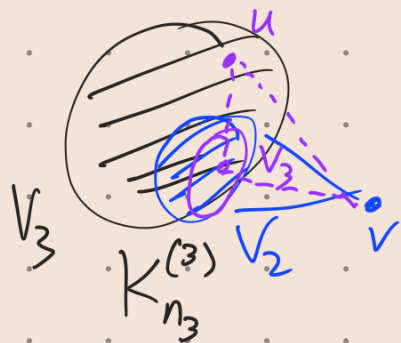
Size of $V_1 \leq V_2$

- $m - \binom{n_3}{3} - \binom{n_2}{2} < \binom{n_2+1}{2} - \binom{n_2}{2} = n_2$

Take all triples $\{u, v, z\}$, $z \in V_1$

$$|\partial\mathcal{F}| = \binom{n_3}{2} + \binom{n_2}{1} + \binom{n_1}{0}$$

$$m = \binom{n_3}{3} + \binom{n_2}{2} + \binom{n_1}{1}$$



This is colexicographic ordering of finite sets of $\mathbb{N}$.

Thm (Kruskal-Katona 63) [Colex min. shadow.]

$\forall \mathcal{F}$ k -unif hypergraph with

$$m = \binom{n_k}{k} + \binom{n_{k-1}}{k-1} + \dots + \binom{n_s}{s}, \text{ then}$$

$$|\partial \mathcal{F}| \geq \binom{n_k}{k-1} + \binom{n_{k-1}}{k-2} + \dots + \binom{n_s}{s-1}.$$

———— Shift / compression ————

Def: $\mathcal{F} \subseteq \binom{[n]}{k}$, $2 \leq i \leq n$, $\forall F \in \mathcal{F}$ define

$$S_i(F) = \begin{cases} F \setminus \{i\} \cup \{1\}, & \text{if } i \in F \text{ \& } 1 \notin F, \text{ and } \\ & F \setminus \{i\} \cup \{1\} \notin \mathcal{F} \\ F & , \text{ otherwise.} \end{cases}$$

Write $S_i(\mathcal{F}) = \{S_i(F) : F \in \mathcal{F}\}$

$\mathcal{F}$ is compressed if $\forall 2 \leq i \leq n, S_i(\mathcal{F}) = \mathcal{F}$.

Fact 2.5 $\mathcal{F} \subseteq \binom{[n]}{k}$, $2 \leq i \leq n$

Map $\mathcal{F} \rightarrow S_i(\mathcal{F})$ is injective $\Rightarrow |\mathcal{F}| = |S_i(\mathcal{F})|$

Fact 2.6 $\partial(S_i(\mathcal{F})) \subseteq S_i(\partial \mathcal{F})$

$$\Rightarrow |\partial(S_i(\mathcal{F}))| \leq |S_i(\partial \mathcal{F})| \stackrel{2.5}{=} |\partial \mathcal{F}|$$

i.e. shifting does not increase shadow size.

Notation Given $\mathcal{F} \subseteq \binom{[n]}{k}$, write

$$\begin{cases} \mathcal{F}_1 = \{F \in \mathcal{F} : 1 \in F\} \\ \mathcal{F}_1^c = \mathcal{F} \setminus \mathcal{F}_1 = \{F \in \mathcal{F} : 1 \notin F\} \\ \mathcal{L}_1 = \{F \setminus \{1\} : F \in \mathcal{F}_1\} \end{cases}$$

Fact 2.7 If $\mathcal{F}$ is compressed, then

$$(i) \quad \partial \mathcal{F}_1^c \subseteq \mathcal{L}_1$$

$$(ii) \quad \partial \mathcal{F} = \mathcal{L}_1 \cup \{E \cup \{1\} : E \in \partial \mathcal{L}_1\}$$

$$\Rightarrow |\partial \mathcal{F}| = |\mathcal{L}_1| + |\partial \mathcal{L}_1|$$

Pf (K.K.) double induction on K then m .